PFAS Comprehensive Overview

PFAS are fluorinated organic chemicals with at least one fully fluorinated carbon atom. PFOS and PFOA are banned; Known as 'forever chemicals' due to strong C-F bonds and persistence.

Why Are PFAS Restricted? AFIRM

Many PFAS have been found to cause long-term health effects depending on the level and duration of exposure, including at very low levels.

PFAS are often referred to as “forever chemicals” because of their resistance to degradation and resulting persistence in the environment, which is a function of the carbon-fluorine bonds they contain (one of the strongest single bonds in chemistry).

Due to their environmental persistence and wide range of hazard traits, authorities around the world are increasingly regulating the entire class of PFAS instead of specific subclasses or individual PFAS.

C8 refers to perfluorooctanoic acid (PFOA), which is a type of per- and polyfluoroalkyl substance (PFAS). PFOA is distinct from perfluorooctane sulfonate (PFOS), which is another type of PFAS.

Both PFOA and PFOS are banned.

C6 including PFHxA and PFHxS

According to the definitions in the latest amendment, PFHxS, its salts and PFHxS-related compounds means:

PFHxS including any of its branched isomers

Its salts

PFHxS-related compounds which are substances that degrade to PFHxS, including substances that contains the chemical moiety C6F13S- as one of its structural elements

PFAS 是含氟有機化合物，至少含一個完全氟化碳原子。PFOS、PFOA 已被禁用，PFAS 被稱為『永久化學品』，因碳氟鍵強，難以降解。

PFAS Introduction

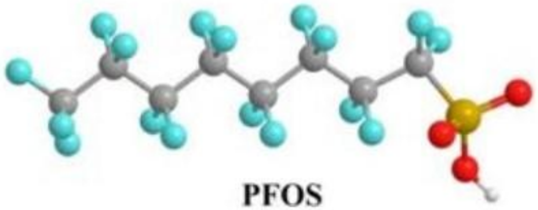
短鏈(Short-chain)				長鏈(Long-chain)				
PFBA (C4)	PFPeA (C5)	PFHxA (C6)	PFHpA (C7)	PFOA (C8)	PFNA (C9)	PFDA (C10)	PFUnA (C11)	PFDoA (C12)

全氟烷基羧酸鹽 (PFCAs)

全氟及多氟烷基物質 (PFAS)

全氟烷基磺酸鹽 (PFSAs)

PFOA

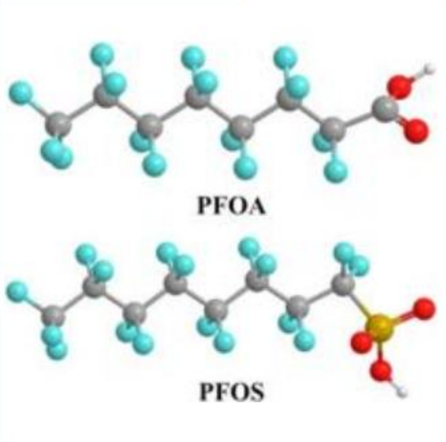


PFOS

化合物	人體半衰期
PFBA (C4)	3天
PFBS (C4)	28天
PFHxS (C6)	5.3年~8.5年
PFOA (C8)	2.1年~3.8年
PFOS (C8)	3年~5年

短鏈(Short-chain)		長鏈(Long-chain)						
PFBS (C4)	PFPeS (C5)	PFHxS (C6)	PFHpS (C7)	PFOS (C8)	PFNS (C9)	PFDS (C10)	PFUnS (C11)	PFDoS (C12)

PFAS – 永久性化學品

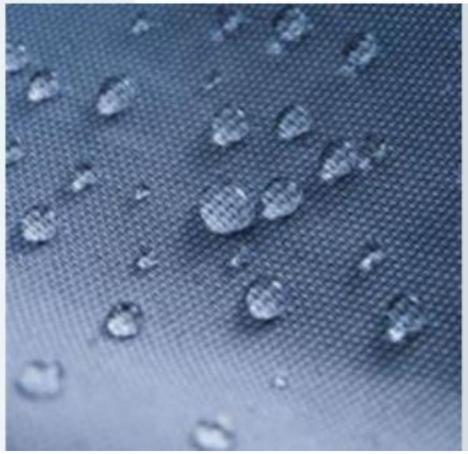


全氟/多氟烷基物質

- 皆為人工合成
- 至少含一個-CF_3或-CF_2的分子

超過15000種物質

- >5000 種全球生產與使用
- 常見PFAS:
  短鏈如 PFHxA、GenX等
  長鏈如 PFOS、PFOA等



防水、防油、高穩定性

- 碳-氟鍵結的穩定結構
- 耐高溫、抗腐蝕

持久性有機汙染物

- PFOA 人體半衰期3~4年
  水中半衰期>92年
- PFOS 人體半衰期3~5年
  水中半衰期>41年

To Know PFAS Better

PFOA

Perfluorooctanoic acid (PFOA) is a specific PFAS compound identified by the Chemical Abstract Services (CAS) Number 335-67-1.
Often referred to as "C8," PFOA is used as an industrial surfactant in chemical processes and as a material feedstock.
PFOA, along with Perfluorooctanesulfonic acid (PFOS), was one of the first PFAS compounds linked to human health effects and persistence in the environment and therefore regulated.

PFOS

PFOS is a specific PFAS compound identified by the CAS Number 1763-23-1.
It is a fluorosurfactant and was previously the key ingredient in various fabric protection products.
PFOS, along with PFOA, was one of the first PFAS compounds linked to human health effects and persistence in the environment and therefore regulated.

PFCs

PFCs Historically referred to Perfluorochemicals or Perfluorinated Compounds, this chemical class is now broadly referred to as "Perfluoroalkyl and Polyfluoroalkyl Substances" or "PFAS" instead.

The term "PFAS" is preferred since "PFCs" is also used in reference to "Perfluorocarbons", which are human made chemicals composed of carbon and fluorine only, and which are regulated due to being powerful greenhouse gases. Perfluorocarbons are separate and distinct from PFAS compounds, with different properties and hazard traits.

PFAS

Perfluoroalkyl and Polyfluoroalkyl Substances (PFAS) are synthetic chemicals defined as "fluorinated substances that contain at least one fully fluorinated methyl or methylene carbon atom (without any H/ Cl/Br/I atom attached to it), i.e., with a few noted exceptions, any chemical with at least a perfluorinated methyl group ($-CF_3$) or a perfluorinated methylene group ($-CF_2-$) is a PFAS."

This definition is provided by the Organization for Economic Co-operation and Development (OECD) which, along with the United States Environmental Protection Agency (U.S EPA), defines several thousand substances as belonging to the group of PFAS.

New legislation in, e.g., California and New York, defines PFAS more broadly as "fluorinated organic chemicals containing at least one fully fluorinated carbon atom."

The challenging part is the current OECD, U.S EPA, and U.S. state definitions are not harmonized.

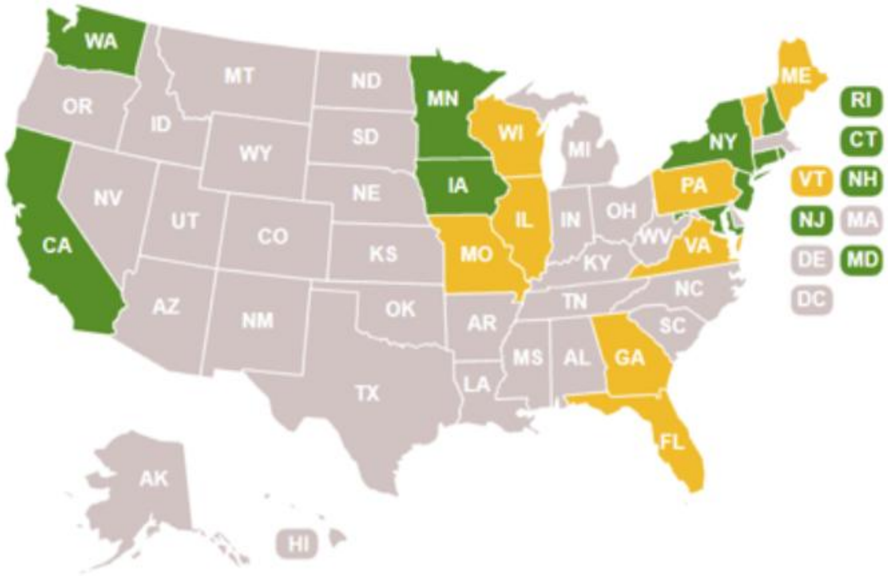
PFAS 美國管制概況與趨勢

目前

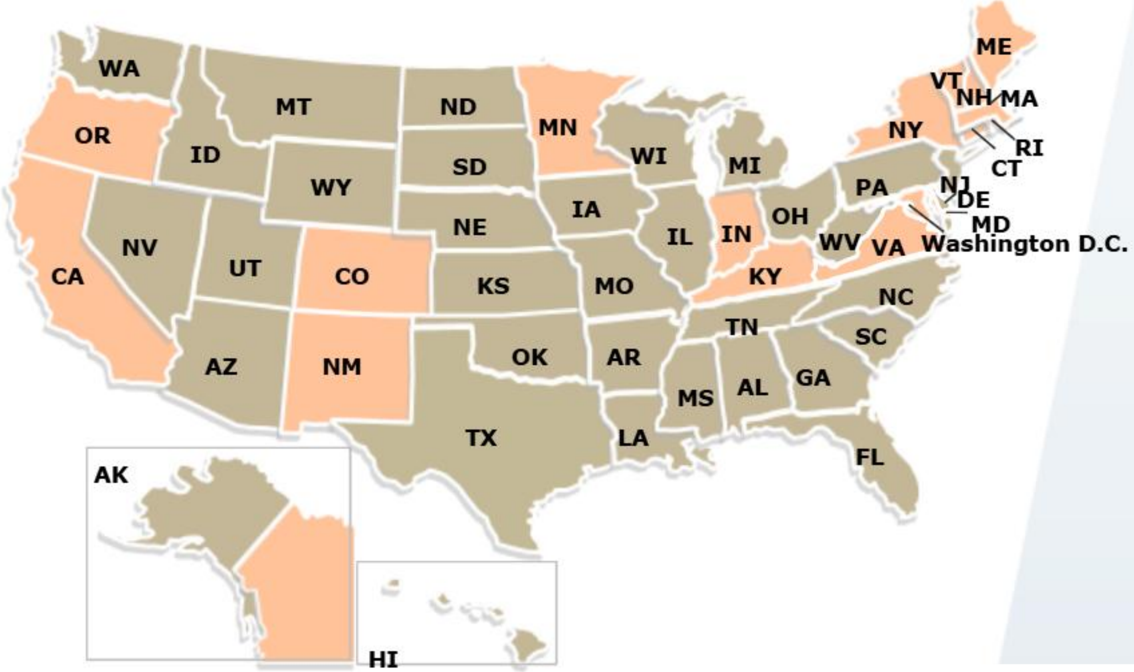
- 美國部分州法對PFAS總量管制 (TOF)
- 少數州有對特定 PFAS 限制
- 目前州法普遍比聯邦嚴格

未來

- 立法限制 PFAS 的州只會增加,產品類型也只會會更多



TPCH



州法管制PFAS

PFAS管制法規 – 美國各州 (部分)

食品包裝

- 加州 AB 1200
- 科羅拉多州 HB 1345
- 緬因州 LD 1433
- 康乃狄克州 SB 837
- 華盛頓州 HB 2658
- 佛蒙特州 S 20 / S 25
- 羅德島 HB 7438 / SB 2044
- 新墨西哥州 HB 212 **New!**

烹飪器具

- 加州 AB 1200 (Labeling)
- 科羅拉多州 HB 1345 (Labeling)
- 緬因州 38 M.R.S. §1614 (Labeling)
- 明尼蘇達州 SF 834 / HF1000
- 佛蒙特州 HB50 (Labeling) / S 0016 / S 25
- 羅德島 HB 5066 (Labeling) / S 0016
- 新墨西哥州 HB 212 **New!**

嬰童用品

- 加州 AB 652
- 科羅拉多州 HB 1345
- 緬因州 38 M.R.S. §1614
- 明尼蘇達州 HF 359
- 羅德島 S.B.2152
- 佛蒙特州 S 25
- 新墨西哥州 HB 212 **New!**

紡織品

- 加州 AB 1817
- 科羅拉多州 HB 1345
- 緬因州 38 M.R.S. §1614
- 明尼蘇達州 HF359
- 佛蒙特州 S20 / S 25
- 羅德島 S.B. 2152
- 新墨西哥州 HB 212 **New!**

消防泡沫

- 加州 SB 1044
- 科羅拉多州 HB 19-1279
- 緬因州 LD 1505
- 康乃狄克州 SB 837
- 明尼蘇達州 HF 359
- 佛蒙特州 S 20 / S 25
- 羅德島 S.B. 2152
- 新墨西哥州 HB 212 **New!**

US California Assembly Bill in SL products

- **California AB 1817** restricts PFAS in textile products, including outdoor apparel, especially those used in wet conditions

- Prohibition of textile articles if PFAS is either intentionally added for a functional or technical effect, or ≥ 100 ppm TOF, from January 1, 2025 (≥ 50 ppm from January 1, 2027)

- California has approved PFAS bans and labeling requirements, with the Department of Toxic Substances Control (DTSC) designated as the enforcement agency. However, the specific enforcement mechanisms have not yet been defined

- All the components used in the textile products, including zippers, rhinestones, etc., are in the scope

加州已經通過 PFAS 禁令和標籤要求: 也就是說, 例如紡織品產品不能含有 PFAS, 並且如果有意添加 PFAS, 需要在標籤上說明.
由加州有毒物質控制部 (DTSC) 負責執法: 這個部門是指定的監管機構, 負責確保企業遵守相關規定, 但具體執法機制尚未明確 - 目前法律已經生效, 但還沒有明確說明:

- 如何檢查產品?
- 違規會有什麼處罰?
- 是否需要提交報告或認證?

簡單來說: 法律已經定了, 但執法細節還沒公布, 企業知道要遵守, 但不知道檢查和罰則怎麼執行!

PFAS管制法規 – 美國各州 (部分)

法規	產品	要求	生效日
加州 65 C.A Prop. 65	All Products 所有產品	PFOA, PFOS, 及PFNA (C9) 若產品含有以上物質需添加警告標籤	已生效 Effected
C.A AB 1817 加州 AB 1817	Textile Products 紡織品	不得有意添加PFAS，或總有機氟小於100 ppm	2025/01/01 Effected
		不得有意添加PFAS，或總有機氟小於 50 ppm	2027/01/01
	Outdoor Apparel 戶外抗水服裝	有意添加PFAS或總有機氟超出 100 ppm需標示	2025/01/01 Effected
		不得有意添加PFAS，或總有機氟小於 50 ppm	2028/01/01

- **Scope: Bans PFAS in apparel and textile items used in wet conditions.**
- **Effective: 2027/01/01 and 2028/01/01**
- **Trigger: Ban applies if PFAS is: • Intentionally added, or • Present at >50 ppm Total Organic Fluorine**
- **Enforcement: Under California DTSC (Department of Toxic Substances Control)**
- **All components included: zippers, rhinestones, etc., must also comply**

美國聯邦 PFAS 管制概況

TSCA Section 8(a)(7) Reporting and Recordkeeping Requirements for Perfluoroalkyl and Polyfluoroalkyl Substances	
調查期間	2011 年 1 月 1 日~ 2022 年 12 月 31 日
常規	申報期間： 2026年4月 13 日 ~ 2026 年 10 月 13 日
小型企業*1	申報期間： 2026年4月 13日 ~ 2027年04月13日

*1根據該規則，小型企業被定義為年總銷售額（與其母公司的總銷售額合計）低於1.2億美元，且化學物質的年產量低於100,000磅的製造商（包括進口商）;或製造商（包括進口商）的年銷售額總額（與其母公司的總銷售額合計）低於1200萬美元。

NATIONAL ARCHIVES

FEDERAL REGISTER

The Daily Journal of the United States Government

Rule

You may be interested in this older document that published on 09/05/2024 with action 'Direct final rule; change to submission period and technical correction.' View Document

Perfluoroalkyl and Polyfluoroalkyl Substances (PFAS) Data Reporting and Recordkeeping Under the Toxic Substances Control Act (TSCA); Change to Submission Period

SUMMARY:

The Environmental Protection Agency (EPA or Agency) is amending the data submission period for the Toxic Substances Control Act (TSCA) PFAS reporting rule by changing the start date for submissions and making corresponding changes to the end dates for the submission period, *i.e.*, the data submission period begins on April 13, 2026, and ends on October 13, 2026, with an alternate end date for small manufacturers reporting exclusively as article importers of April 13, 2027. As promulgated in October 2023, the regulation requires manufacturers (including importers) of perfluoroalkyl and polyfluoroalkyl substances (PFAS) in any year between 2011-2022 to report certain data to EPA related to exposure and environmental and health effects. This change is necessary because EPA requires more time to prepare the reporting application to collect this data. The Agency is separately considering reopening certain aspects of the rule to public comment. The delayed reporting date ensures that EPA has adequate time to consider the public comments and propose and finalize any modifications to the rule before the submission period begins.

DATES:

This interim final rule is effective on May 13, 2025. Comments must be received on or before June 12, 2025.

美國各州 PFAS 管制概況

TPCH (禁用)	
PFAS	< 100 ppm TOF/TF
鉛、鎘、汞與六價鉻	< 100 ppm
鄰苯二甲酸酯 (可塑劑)	< 100 ppm

州聯合協議(19州): TPCH 美國包裝材料示範法案

2021更新 : 新增 PFAS和鄰苯二甲酸酯

TOXICS IN PACKAGING CLEARINGHOUSE

PFAS (Per- and polyfluoroalkyl substances) Test Method Updates

- Important Update on PFAS Testing: EN 17681-1:2025 Standard
- It will likely become the dominant test method for textile products in the global market.

	Current	Updated
Test Method	EN 17681-1/2:2022	EN 17681-1:2025
Extraction Method	Methanol extraction method 僅使用甲醇萃取	Methanol extraction and alkaline hydrolysis 結合甲醇萃取與鹼性水解
Strength and Limitation	Not sufficient to extract and detect side-chain fluorinated polymers that are included in EU regulatory restrictions if they are PFOA, C9-C14 PFCA and PFHxA-related compounds 無法有效檢出 含側鏈氟聚合物，例如：PFOA、C9–C14 PFCA、PFHxA 等	Can detect more PFAS content by “cutting” side chain from PFAS polymers; Some PFAS analytes (e.g. CAS 27854-31-5, 2043-54-1, 30046-31-2) will not be detected due to their chemical properties; They need to be detected by EN 17681-1/2:2022 可透過“切斷側鏈”檢出更多 PFAS 多聚物中的殘留量，但某些 PFAS因化學特性仍無法偵測，需回頭用 2022 版檢測
Effective Date	Current	Published on April 30, 2025
Detection Accuracy	Normal	Significantly higher

vuori

EU updates on PFAS (per- and polyfluoroalkyl substances)

- Groups restricted under EU POPs regulation (PFOS, PFOA, PFHxS)
- Group restricted in EU REACH (C9-C14 PFCAs)

- **Upcoming October 2026 EU REACH restriction** on PFHxA (undecafluorohexanoic acid) – Annex XVII Entry 79

- Current EU proposal to restrict all PFAS – a record number of scientific and technical comments were received (> 5600). The European Chemicals Agency (ECHA) will consider all comments, continue evaluation and form opinions in 2025. In 'opinion development' stage

市場案例與風險 | Market Cases & Risks

•Recall Alert (SR/00537/25): •Product: Hooded jacket •Country: UK origin, recalled in Belgium •Reason: Shell fabric contained excessive PFAS. •Action Taken: Product withdrawn from the market

Product recalls and settlements

EU Safety Gate

First PFAS recall of apparel

- Alert No. : SR/00537/25
- Published: February 6, 2025
- Notifying country: Belgium
- Country of origin: United Kingdom
- Product: Hooded jacket
- Failed component: Shell fabric
- Failure results:
  - The product contains an excessive amount of perfluorooctanoic acid (PFOA) (measured value: 34 µg/kg)
  - The product does not comply with the Persistent Organic Pollutants (POP) Regulation
- Follow-up action: Withdrawal of the product from the market

https://ec.europa.eu/safety-gate-alerts/screen/webReport/alertDetail/10092891?lang=en

市場案例與風險 | Market Cases & Risks

Product recalls and settlement

EU Safety Gate

First PFAS recall of home textile

- Alert No. : SR/00555/25
- Published: February 6, 2025
- Notifying country: Belgium
- Country of origin: Belgium
- Product: Mattress protector
- Failed component: Shell fabric
- Failure results:
  - The product contains an excessive concentration of 2-(N-ethylperfluorooctane-1-sulphonamido)-ethanol (N-Et-FOSE) (measured value up to: 2.3 μg/m^2). This substance is carcinogenic and poses a risk to the environment
  - The product does not comply with the Persistent Organic Pollutants (POP) Regulation
- Follow-up action: Withdrawal of the product from the market

https://ec.europa.eu/safety-gate-alerts/screen/webReport/alertDetail/10093020?lang=en

•Alert No.: SR/00555/25 (Safety Gate) •Notified by: Belgium •Product: Mattress protector (origin: Belgium) •Issue: Excessive PFAS in shell fabric (substance is carcinogenic and environmentally hazardous) •Action Taken: Product withdrawn from the market

Update on PFAS - Regulation

- Restrictions in global markets 全球市場的限制現況
  - EU: POPs, REACH
  - US: TPCH, state laws e.g., California, Washington, Maine, New York, etc.
  - Asia Pacific countries: Japan, Korea, China, Taiwan, Indonesia, etc.

- Proposed regulations 法規草案與未來趨勢
  - Many bills at the US state level proposed over the last few years to restrict/prohibit certain consumer products containing PFAS, along with implementing requirements for reporting PFAS usage.
  - Feb 2023 – EU – Denmark, Germany, the Netherlands, Norway and Sweden submitted a restriction proposal to ECHA covering a wide range of PFAS uses. Six-month consultation period opened in March 2023

- High concern about the usage of PFAS in the market 市場關注度極高

Update on PFAS - EU and Other (European) Countries

- Existing Regulatory Requirement

	EU REACH / POPs (EU & EEA Countries)	Switzerland ORRChem	Turkey POPs
Perflurooctane sulfonates (PFOS) and its derivatives	Textile/ Coated material: 1 μg/m²; Semi-finished products or articles: 0.1% Substance or constituent of preparations: 10 mg/kg	Textile/ Coated material: 1.0 μg/m²; Others: 0.1%	Textile/ Coated material: 1 μg/m²; Semi-finished products or articles: : 0.1% Substance or constituent of preparations: 10 mg/kg
Perflurooctanoic acid (PFOA) and its salts	25 ppb (0.025 mg/kg) (Norway: Textile / Carpets / Coated material: 1 μg/m²)	25 ppb (0.025 mg/kg)	/
Perflurooctanoic acid (PFOA) and its derivatives	1000 ppb (1 mg/kg) (each or total)	1000 ppb (1 mg/kg) (each or total)	/
C9-C14 Perfluorocarboxylic acids (PFCA) and their salts	25 ppb (0.025 mg/kg) (total) [Feb 25, 2023]	25 ppb (0.025 mg/kg) (total) [Oct 1, 2022]	/
C9-C14 Perfluorocarboxylic acids (PFCA) and their derivatives	260 ppb (0.26 mg/kg) (total) [Feb 25, 2023]	260 ppb (0.26 mg/kg) (total) [Oct 1, 2022]	/
Perfluorohexane sulfonic acid (PFHxS) and its salts	25 ppb (0.025 mg/kg) [28 Aug 2023]	25 ppb (0.025 mg/kg) (total) [Oct 1, 2022]	/
Perfluorohexane sulfonic acid (PFHxS) and its derivatives	1000 ppb (1 mg/kg) (total) [28 Aug 2023]	1000 ppb (1 mg/kg) (each or total) [Oct 1, 2022]	/

Information updated on June 2024

Update on PFAS - EU Restriction Proposal

Restriction Option (RO) for PFAS	Scope	Requirement	Remark
RO1	• Full ban	Prohibited	• Enters into force after a transition period of 18 months • No derogations
RO2	• Substances on their own	Prohibited	• Enters into force after a transition period of 18 months • 5 or 12 years for derogations after the transition period ends • Time-unlimited derogations only for specific uses
	• Constituent of another substance • Mixtures • Articles	≤ 25 ppb for any targeted PFAS (polymeric PFAS are exempt) ≤ 250 ppb for sum of targeted PFAS (polymeric PFAS are exempt) ≤ 50 ppm for PFAS, including polymeric PFAS*	

•March 22, 2023 – start of six-month consultation
•2024 – opinion of Committees for Risk Assessment (RAC) and Social-economic Analysis (SEAC)
•2025 – Commission decision and entry into force
•2026/2027 – restriction becomes effective

Update on PFAS - United States

- Over the years, several U.S. states have implemented restrictions on the distribution or sale of products containing PFAS, with a number of bills introduced at the state level to restrict or prohibit PFAS in various products:

  - California - several state laws restrict PFAS in different product categories: AB 1200 (plant-based food packaging and cookware), AB 652 (juvenile Products), AB 1817 (apparel and textile products), AB 2762 (cosmetics)
  - New York - S1322/A994 prohibition on intentionally added PFAS in apparel, starting in 2025. After January 1, 2028, a new restriction will go into effect that applies specifically to outdoor apparel for severe wet conditions with intentionally added PFAS.
  - Minnesota - As of January 1, 2025, Minn. Stat. §116.943 prohibits 11 categories of products containing intentionally added PFAS. These include carpets or rugs, fabric treatments, textile furnishings, and more.
  - Colorado – On May 1, 2024, the governor of Colorado signed SB 24-081 into law to revise the state's aforementioned act.
    - Revokes the 2025 prohibition of outdoor apparel for severe wet conditions containing intentionally added PFAS, unless these products are accompanied by the phrase 'Made with PFAS Chemicals', on January 1, 2028, but will prohibit these goods from January 1, 2028

Update of PFAS - United States

- Maine - amended its law (38 MRSA §1614) on products containing intentionally added PFAS and will be implemented in phases, starting in January 2026.
  - directs manufacturers to notify the Department of Environmental Protection (DEP) by January 1, 2025, if their products contain intentionally added PFAS.

  通報義務：「2025 年開始，產品中若有 PFAS，製造商要通報給環保局（DEP）。」
  - bans carpets, rugs and fabric treatments from January 1, 2023, if these contain intentionally added PFAS.

  禁止範圍擴張：「從 2023 年就已禁止地毯、布料處理劑中含有 PFAS。」
  - products containing intentionally added PFAS will be prohibited from January 1, 2030, unless the DEP has determined by rule that the use of PFAS is unavoidable.

  最終禁用目標（2030）：「除非 DEP 認定該產品無法避免使用 PFAS，否則 2030 年起全部禁止。」

- In the US, a host of jurisdictions across the nation have implemented measures to regulate PFAS chemicals in consumer goods.

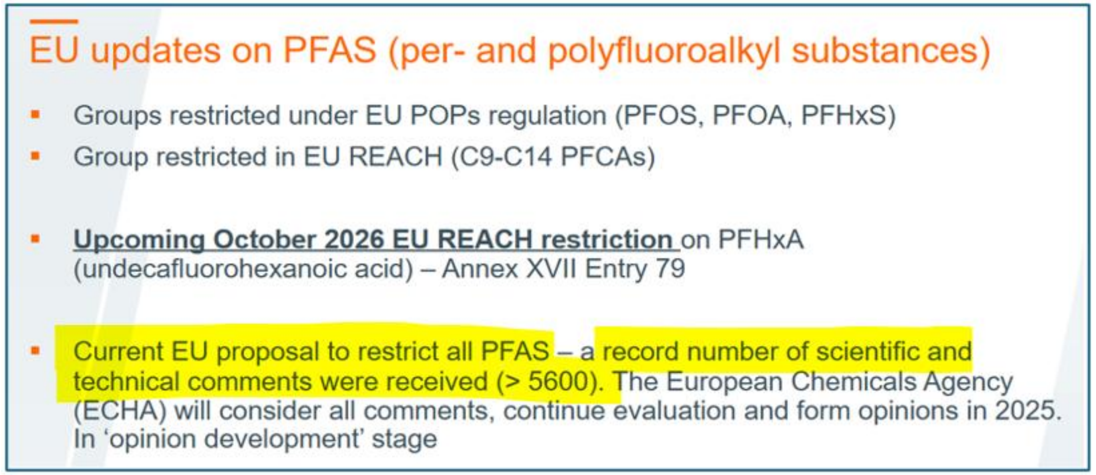
PFAS Free? – Not recommended

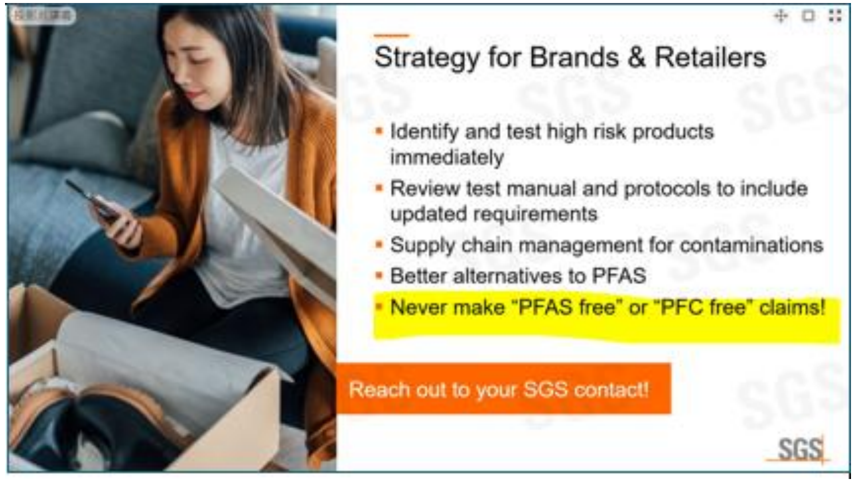
PFAS substances are still evolving and will likely become similar to SVHC (Substances of Very High Concern) under REACH

Future regulations may broaden the definition

Trace contamination or unintentional presence could invalidate the claim

Industry best practice (SGS guidance) advises against such claims





PFAS compliance – recommended statements:

- Vuori does not intentionally use PFAS chemistry in our products
- For recent seasons, PFAS testing has shown non-detect results for regulated PFAS substances using EN 17681-1:2022 and EN 17681-2:2022 methods
- Please note that PFAS definitions and restrictions under EU REACH and POPs are evolving, and industry guidance recommends avoiding absolute claims such as "PFAS Free"
- Our approach is to maintain compliance with current regulations and continuously monitor updates to ensure responsible product stewardship